

STEAMBOAT SPRINGS, CO, USA

WMO: 724678

Lat: 40.517N

Lon: 106.867W

Elev: 2097

StdP: 78.55

Time Zone: -7.00 (NAM)

Period: 11-19

WBAN: 94033

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-18.5	-24.8	0.5	-17.0	-21.7	0.7	-13.2	8.3	-2.8	7.2	-2.9	0.1	100	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	30.6	12.8	29.3	12.7	28.0	12.5	15.6	24.6	14.9	24.2	14.1	23.4	3.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	12.3	16.2	11.9	11.3	16.4	10.8	10.4	16.2	51.5	24.9	49.0	23.9	46.7	23.4	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.2	5.6	DB	-27.4	32.1	3.6	1.7	-30.0	33.4	-32.1	34.4	-34.2	35.4	-36.8	36.6	(4)
(5)			WB	-27.7	16.1	3.6	0.7	-30.2	16.6	-32.3	17.0	-34.3	17.4	-36.9	17.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.7	-7.6	-4.5	0.6	4.8	9.4	15.6	18.9	17.6	13.7	6.1	-0.3	-7.2		
	DBStd	9.99	5.23	4.66	4.44	3.70	3.61	3.09	1.90	1.98	3.19	4.71	4.66	5.28		
	HDD10.0	2439	544	406	293	159	56	3	0	0	8	130	308	532		
	HDD18.3	4687	803	639	551	406	278	90	16	37	141	378	558	790		
	CDD10.0	851	0	0	1	3	35	171	277	235	119	10	0	0		
	CDD18.3	60	0	0	0	0	0	8	35	14	3	0	0	0		
	CDH23.3	2129	0	0	0	1	18	481	850	593	186	0	0	0		
	CDH26.7	594	0	0	0	0	1	132	272	157	32	0	0	0		
(14) Wind	WSAvg	1.4	0.6	1.1	1.3	2.0	2.0	1.8	1.6	1.6	1.5	1.4	1.0	0.7		
(15) Precipitation	PrecAvg	575	61	50	47	57	52	39	36	40	47	44	54	59		
	PrecMax	902	149	119	82	104	144	142	83	90	194	102	142	161		
	PrecMin	407	16	20	14	14	8	2	3	10	6	2	13	15		
	PrecStd	115	33	22	19	23	27	28	20	21	37	26	27	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.5	8.8	17.9	21.6	25.6	32.0	32.3	31.2	29.9	22.5	16.7	6.8		
		MCWB	0.2	2.6	5.5	6.4	8.9	12.0	12.4	12.4	12.7	9.1	5.5	1.1		
	2%	DB	3.2	6.1	14.2	18.7	23.3	29.9	30.7	29.7	27.4	20.4	13.2	4.1		
		MCWB	-0.4	1.0	4.3	6.4	8.5	12.0	13.2	12.5	11.3	8.0	4.3	0.0		
	5%	DB	1.3	4.3	11.2	16.2	21.4	28.0	29.6	28.6	25.4	18.5	10.5	1.7		
		MCWB	-1.1	0.4	3.1	5.4	8.2	11.8	13.2	12.3	10.5	7.3	3.1	-1.3		
	10%	DB	-0.1	2.6	8.6	13.5	19.2	26.4	28.3	27.2	23.8	16.0	7.9	0.1		
		MCWB	-1.7	-0.3	1.9	4.6	8.0	11.2	13.1	12.3	10.4	6.5	2.0	-2.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	3.2	6.0	8.0	11.0	15.1	17.0	16.2	13.9	11.2	6.4	2.2		
		MCDB	3.1	7.0	16.5	18.1	21.4	27.2	24.3	25.0	24.8	18.7	13.9	6.4		
	2%	WB	0.4	1.6	4.9	6.9	10.0	14.1	15.8	15.3	13.0	9.7	4.9	0.5		
		MCDB	2.2	4.5	13.3	16.5	20.0	25.3	24.3	24.3	22.4	17.7	12.0	2.9		
	5%	WB	-0.5	0.8	3.5	6.2	9.1	13.2	15.2	14.4	12.3	8.3	3.7	-0.6		
		MCDB	0.9	3.4	10.2	14.3	18.7	24.1	24.3	23.0	21.3	16.0	9.8	1.3		
	10%	WB	-1.8	0.1	2.3	5.1	8.4	12.3	14.6	13.5	11.6	7.1	2.4	-1.8		
		MCDB	-0.1	2.4	7.7	12.3	17.4	23.4	23.7	22.3	20.1	14.7	7.2	-0.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	10.8	12.8	12.9	14.8	19.3	18.2	18.5	17.9	14.9	12.7	10.5		
		MCDBR	12.0	11.9	16.4	18.5	19.7	22.1	20.6	21.3	21.3	20.1	16.9	12.2		
	5% WB	MCWBR	8.5	7.4	8.0	8.1	7.7	8.0	6.6	7.1	8.1	9.2	8.9	8.5		
		MCWBR	9.1	9.9	15.0	16.1	17.2	19.6	17.2	17.2	17.1	16.8	15.2	10.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.227	0.248	0.263	0.316	0.322	0.332	0.359	0.347	0.302	0.266	0.244	0.235		
		taud	2.394	2.310	2.363	2.355	2.453	2.422	2.398	2.427	2.562	2.641	2.558	2.420		
	Ebn at Noon	Ebn at Noon	962	980	998	957	954	939	910	914	945	957	940	922		
		Edh at Noon	79	102	109	119	111	115	116	110	89	73	68	72		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.23	3.08	4.49	5.50	6.24	7.50	6.97	5.98	5.12	3.60	2.43	1.85		
	RadStd	RadStd	0.13	0.22	0.39	0.41	0.46	0.46	0.30	0.28	0.34	0.31	0.26	0.16		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page